

Neeraj Jadhav

Education

- 2024-26 **Master of Science**, *Computer Science*, Specialization: Computing Systems
Master's Thesis: ML-aided Code Optimization
Prof. Santosh Pande, [Georgia Institute of Technology](#), GPA: 3.80
Key courses: High Perf. Computer Arch., Adv. Compilers, GPU HW & SW, Secure Computer Sys.
- 2020-24 **Bachelor of Technology with Honors**, *Computer Science and Engineering*
Minor in Artificial Intelligence and Data Science
[Indian Institute of Technology, Bombay](#), (Major) CGPA: 9.34, (Minor) CGPA: 9
Key courses: Algorithms, Operating Systems, Compilers, Optimization Models, AI, and ML

Publications

- **Phaedrus: Exploring Dynamic Application Behavior with Lightweight Generative Models and Large-Language Models** (OOPSLA '26)
Bodhisatwa Chatterjee, **Neeraj Jadhav**, and Santosh Pande

Honors & Awards

- Recipient of **Outstanding Master's Research Award** by [College of Computing, GT](#) ['25]
- Recipient of **Master's Graduate Fellowship in CS** by [Georgia Institute of Technology](#) ['24]
- Amongst **top 10** teams in **Algorithmic Trading** contest organized by Web and Coding Club ['22]
- Secured **46th**/150k and **186th**/870k national ranks in **JEE Advanced** and **Main** exams resp. ['20]

Experience

- Summer '25 **ML Compilers Internship**, [Qualcomm Inc.](#), Boxborough, MA
○ Optimized **ML compiler for devices**, improving modularity, and **reducing compilation time** by $\sim 3\times$
○ Implemented **quantization** and **LLM optimizations** for efficient inference on **Qualcomm accelerators**
○ Improved **cross-team collab** by deliverables in: Parsing, IR modification, **CodeGen**, **Perf. Analysis**
- Summer '23 **Quantitative Research Internship**, [Quadeye Securities LLP](#), Gurugram, India
○ Implemented a **trading simulator** and optimization scripts in **R** to emulate **high-frequency trading**
○ **Optimized** the strategy using **black-box methods** improving **time spent** by **80%** and **PnL** by **2%**
- Winter '22 **Software Engineering Internship**, [FinIQ Consulting](#), Pune, India
○ **Enhanced** the proprietary financial coding language **FinSPL**, by adding **operational parallelism** and **efficient** data structures for information **storage** and **retrieval** of streaming **market indices**
- Summer '22 **Automated Orchestration using LLMs**, [IBM Research](#), Gurgaon, India
○ **Streamlined** business task execution using **LLMs** and **process mining** to optimize **workflow efficiency**
○ Pipelined **data extraction** from **conversation logs** by fine-tuning **GPT2** for seamless **API calls**

Research

- Fall '24 **ML-aided Program Analysis and Optimization**, *Master's Thesis*
Prof. Santosh Pande ([Georgia Tech](#))
○ Analysed **SPEC** benchmarks with **GPT-4**; **>10%** reduction in binary size via **compiler optimizations**
○ Leveraged LLMs with **profile-guided optimization (PGO)** to accurately predict expensive functions with **85%** execution coverage, reducing software-profiling overhead and improving runtime by **$\sim 10\%$**
- Fall '23 **Robust Celltree : A Paradigm for Distributed Data Repositories**, *Bachelor's Thesis*
Prof. Manoj Prabhakaran ([IIT Bombay](#)), *Prof. Indranil Gupta* ([UIUC](#))
○ Designed a **new data structure** with **safety** guarantees to **minimize** computational **load and time**
○ Enhanced **blockchain** with **proof-of-stake** and modified **consensus**, achieving high **latency gains**
- Fall '23 **mpC : A Secure Multiparty Computation Language**
Prof. Manoj Prabhakaran ([IIT Bombay](#))
○ Developed a **language**, with backend in **C++**, for **abstracting** out **cryptographic protocols**
○ Secured **communication with TLS** & **reduced** compilation **time** for **rapid testing** of new protocols

Projects

- Fall '25 **GPU HW & SW Optimizations**
○ Optimized **CUDA kernels** for tiled **matrix multiplication** and bitonic sort using shared & pinned memory, **tiling**, loop unrolling, **kernel fusion & fission**, async operations, **memory transfer**, etc.
○ Implementing **CUDA kernels for Softmax, GEMM, and Multi-head attention** to achieve an **LLM inference pipeline** having **FlashAttention2** support, optimized with **KV-cache** for token times

- Spring '25 **Array Bounds Check Optimizer**
- Developed **LLVM compiler passes** to eliminate **redundant checks** and improve runtime by **~20%**
 - Applied **Gupta's algorithm** for redundancy removal via **check splitting** and **selective elimination**
 - Implemented **ABCD algorithm** for **JIT** applications and check removal by **graph-based algorithms**
- Spring '25 **Securing Computer Systems**
- Implemented a **trusted computing base (TCB)** and **memory protection**, reducing flaws by **40%**
 - Designed and deployed **authentication and access control**, improving security enforcement by **35%**
- Fall '24 **Benchmarking Erasure codes**
- Designed a **pipeline using Python** for evaluating the **performance** of **Raptor and LT** erasure codes
 - **Simulated** various transmission channels with the **aff3ct** library to accurately emulate **packet loss**
- Spring '24 **Market Index Data Analysis**
- Developed **linear regression** models for 15 market **indicators** to **analyze** underlying hidden **features**
 - Performed **feature engineering** using **trading insights** to **eliminate** several **hyperparameters**
- Spring '24 **Portfolio Optimization**
- Developed a **Python simulator** to efficiently **evaluate** and analyze **Markowitz portfolio optimization**
- Fall '23 **Graph Neural Networks for Optimization**
- Modeled **linear programs** as graphs to solve using **GNNs**, achieving fast and **efficient optimization**
 - Solved a **jigsaw puzzle** using **LP** and **GNN**, achieving a **5x speed-up** over **traditional LP solvers**
- Fall '23 **Abstract Protocol Design**
- Built a **verifiable secret sharing** scheme in **self-built** language **mpC**, eliminating formal verification
- Spring '23 **Blockchain simulator**
- Developed a **blockchain** using **SymPy in Python** to analyze **selfish-mining** and **related attacks**
- Spring '23 **Signal Transformation Tool**
- Designed a **signal processing** tool with **PyQt** and **Tkinter** for **teaching embedded systems**
- Spring '23 **Optimal Data Subset Selection for Model Training**
- **Optimized model training** by **scoring datapoints** to eliminate data **redundancy** over **75%**
- Spring '23 **GCC-like Compiler**
- **Implemented a compiler** for a C-like language **from scratch** using **Flex, Bison and C++**
- Spring '23 **Postgres++**
- **Enhanced PostgreSQL** by **adding functional dependency** support to **simplify complex schemas**
- Fall '22 **Efficient Small-scale Face Detector**
- **Developed a face detector** from scratch using **HoG in Python**, achieving **~2x SOTA** runtimes
- Fall '22 **Custom Animation Rendering Engine**
- **Built a 3D animated world** from scratch in **OpenGL (C++)**, awarded **Best Course Project**
- Spring '22 **Facial Recognition Internship**
- Developed **face recognition** pipelines with **OpenCV, Azure, and AWS APIs** for **attendance tools**
- Spring '22 **Peer-to-Peer File Sharing App**
- Developed a **reliable P2P network** in **C++** with error handling to ensure **lossless data sharing**
- Spring '22 **RISC in VHDL**
- Implemented a **finite state automaton** in **VHDL**, achieving a functional **CPU computation pipeline**
- Spring '22 **Speech synthesizer**
- **Trained HMMs with Kaldi** for **speech-to-text** transcription, **automating the manual work**
 - Developed **text-to-speech synthesizers** with **SoX and Python-Coqui** for **Marathi and Kannada**
- Fall '21 **Learning Management System Web-App**
- **Built a remote learning web app** using **Django, SQLite, and React** for **fast resource access**
 - Created a **Bash CLI tool** to handle **HTTP and TCP requests** as a **browser interface alternative**
- Fall '21 **Rapid Visualization of Fractals**
- Developed a **Mandelbrot set visualizer** using **SDL in C++**, enabling **efficient fractal exploration**
 - **Optimized** the visualizer tool with **BSTs, Heaps, and KDTrees** for **~10ms smooth magnification**
- Fall '21 **Generative Pre-training of Pixels**
- Integrated **image generation** into **GPT-2** using **vectorization**, eliminating training of **GANs**
- Summer '21 **Computer Vision Basics**
- Built **CNNs and Vanilla NNs** in **PyTorch** for **MNIST digit classification** with **98% accuracy**
- Fall '20 **Visual Maze solver**
- **Solved a visual maze** with **grid extraction and graph algorithms** in **~50ms** using **Python**

Fall '20 **Game Development**

- Developed a **coin-catch game** in **C++** using the **Turtlesim graphics** library at IIT Bombay
- Built an **alien shooter** game in **PyGame** collecting **responses** from **~50 users** for **bot** development

Involvement

Mentorships **Tech Mentor at XLR8 Robotics competition**, *IIT Bombay*

TAs **Calculus & Embedded Systems**, *IIT Bombay*, **Algorithms**, *GATech*

Skills

Languages

Proficient: C/C++, Python || Familiar: LLVM IR, Torch ATen IR, CUDA, R, Rust, Go, VHDL

Libraries

PyTorch, Tensorflow, Huggingface, LangChain Django, OpenCV, PyAutoGUI, Pandas, Scikit-Learn

Softwares

Git, MATLAB, Wireshark, PostgreSQL, React.js, Node.js, Android Studio, AWS, Azure, Docker